Python Collections

Set

A **set** in Python is a collection that has special properties:

- **Unordered** → Items do not have a fixed position (no indexing).
- **Mutable** → We can add or remove items (but cannot access or change by index).
- **No Duplicates Allowed** → Duplicate values are automatically removed.
- Written with **curly braces { }**.

Creating a Set

```
myset1 = {10, 20, 30, 40, 50}
myset2 = {"apple", "mango", "cherry"}
myset3 = {10, "apple", "A", True}  # can store mixed data types

myset4 = set()   # empty set (use set(), {} creates a dictionary)
```

Accessing Set Items

- You **cannot access items by index** (since sets are unordered).
- You **cannot update specific items**.
- But you can **loop through** the set.

```
myset = {"apple", "banana", "cherry"}
for x in myset:
    print(x)
```

Check if Item Exists

```
myset = {"apple", "banana", "cherry"}
print("banana" in myset)   # True
print("orange" in myset)   # False
```

```
if "banana" in myset:
    print("Yes, banana exists")
```

Set Functions

```
myset = {"apple", "banana", "cherry"}
print(len(myset))   # 3 (size of the set)
```

❌ Counting duplicates → **Not possible** (since duplicates don't exist).
❌ Sorting / Reversing → **Not possible** (since set is unordered).

Adding Items

- **add()** → adds a single item
- **update()** → adds multiple items

```
myset = {"apple", "banana", "cherry"}
myset.add("orange")
print(myset)  # {'banana', 'apple', 'orange', 'cherry'}

myset.update(["mango", "grapes"])
print(myset)  # {'banana', 'apple', 'cherry', 'grapes', 'orange', 'mango'}
```

Removing Items

1. **remove(value)** → removes item, error if not found.
2. **discard(value)** → removes item, no error if not found.
3. **pop()** → removes a random item.

```
myset = {"apple", "banana", "cherry"}
myset.remove("banana")
print(myset)  # {'apple', 'cherry'}
myset.discard("xyz")  # No error even if item not present
myset.pop()  # removes random item
print(myset)
```

Clear / Delete a Set

```
myset = {"apple", "banana", "cherry"}
myset.clear()
print(myset)   # set() → empty set

del myset
# print(myset)  # NameError (set is deleted)
```

Copying a Set

```
myset1 = {"apple", "banana", "cherry"}

myset2 = myset1.copy()       # Approach 1
myset3 = set(myset1)         # Approach 2

print(myset2)
print(myset3)
```

Joining Two Sets

1. Union → Combines all items from both sets

```
set1 = {"a", "b", "c"}
set2 = {1, 2, 3}
set3 = set1.union(set2)
print(set3)  # {'a', 'b', 'c', 1, 2, 3}
```

2. Intersection → Returns only common items

```
set1 = {"apple", "banana", "cherry"}
set2 = {"google", "microsoft", "apple"}

set3 = set1.intersection(set2)
print(set3)  # {'apple'}
```

Summary

- **Set is Unordered** → No indexing, sorting, or reversing.
- **Mutable** → You can add or remove items.
- **No Duplicates** → Each item is unique.